

Controllable multimodal generation

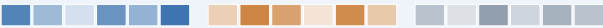
REQUESTED SCENE

A woman in a red coat gives a blue cup to a man in a green sweater.
A child beside them reads a book.

2.1 Learn representations and component extractors

Learn from fixed encoders or downstream tasks / rewards.

Illustrative: participants / roles / other



2.2 Jointly learn denoising order, component dependencies, and the transformer

Dependency diagnostics

Compare noise conditions at the current model checkpoint.



Schedule + transformer updates

Dependencies and denoising order evolve together with training.

Repeat diagnostics as training changes the model.



2.3 Shared and specialized transformer computations

Noise-dependent module gates



Sparse attention

Shared / switched MLPs

Goal: accurately generate participants, attributes, and relationships.